LLAMA Case-study problems:

Counting params / FLOPs.

1) Attn params = $\underbrace{DNH + DKH + DKH + NHD}_{\text{projs}} + \underbrace{D}_{\text{layer-norm}}$

$$MLP = 2 \cdot DF + FD$$

$$\text{Total} = 2DV + (2DNH + 2DKH + D + 3FD) \cdot L$$

$$= 2 \cdot 8192 \cdot 128256 + (2 \cdot 8192 \cdot 64 \cdot 128 + 2 \cdot 8192 \cdot 8 \cdot 128 + 8192 + 3 \cdot 28672 \cdot 8192) \cdot 80$$

$$= 7 \cdot 10^{10} = 70B \text{ params}$$

$$\text{Total}_{\text{Attn}} \simeq 12 \cdot 10^9 = 12B \text{ params}$$

$$\text{Total}_{\text{MLP}} \simeq 56.4 \cdot 10^9 = 56B \text{ params}$$

$$\text{Total}_{\text{vocab}} \simeq 2B \text{ params}$$

Note: MLP params account for most of model's weights so we can ignore all other params when reasoning about memory / FLOPs. (can use simplified transformer layer from prev section)

2) $\text{FLOPS/}_{\text{token}} = 6 \cdot 70B = 420 \cdot 10^9$ FLOPS

If we're compute bound + MFU = 1 in bf16 on 1 TPU:

$$T_{\text{comp}} = \frac{420 \cdot 10^9 \text{ FLOPS}}{4.59 \cdot 10^{14} \text{ FLOPS/s}} \simeq 10 \cdot 10^{-4} = 1\text{ms}$$

3) $FLOPS_{Total} = 6 \cdot 70B \cdot 15T = 6 \cdot 70 \cdot 10^9 \cdot 15 \cdot 10^{12}$

$$= 6.3 \cdot 10^{24}$$

Assuming compute bound w/ MFU = 1, bf16 on a single v5p:

$$T_{comp} = \frac{6.3 \cdot 10^{24}}{4.59 \cdot 10^{14}} = 1.37 \cdot 10^{10} s$$

4) $$T_{comp} = \frac{6.3 \cdot 10^{24}}{8960 \cdot 4.59 \cdot 10^{14} \cdot 0.4} \simeq 3.8 \cdot 10^6 s$$

5) $Bytes_{params} = 70B \cdot 2 = 140 \cdot 10^9$

$Bytes_{opt} = 2 \cdot 4 \cdot 70B = 560 \cdot 10^9$

(Assuming each checkpt is of activations size)

$Bytes_{chckpt} = 2 \cdot B \cdot D \cdot L = 2 \cdot 4 \cdot 10^6 \cdot 8192 \cdot 80 \cdot 4$

$= 5.24 \cdot 10^{12} \cdot 4 \simeq 20.9\ TB$

$Total = 140 GB + 560 GB + 20.9 TB = 21.6 TB$

$$\#\ TPUs, min = \frac{21.6 \cdot 10^3\ \cancel{GB}}{96\ G\cancel{B}} \simeq 225\ TPUs$$

6) $$Bytes/chip = \frac{21.6 \cdot 10^{12}}{8960} = 2.4 \cdot 10^9 = 2.4 GB$$

Sharding LLaMA 3-70B for training:

7) With no context / sequence parallel, we can only utilize $4 \cdot 10^6 / 4096 \simeq 976$ chips. Pure FSDP is not possible here.

We could technically still do it over 976 chips only but it would be an waste for the rest..

8) $BS_{Local} = 4 \cdot 10^6 / 8960 \simeq 446$

$$\frac{C}{3 \cdot W_{ici}} = \frac{2550}{3} \simeq 833$$

$446 < 833$ so we would be comm-bounded.

FSDP only is still not a good idea.

9) $$\frac{\alpha^2}{M_x M_y F} = \frac{(2550)^2}{2 \cdot 28672} \simeq 113$$

$\frac{B}{N} = 446 > 113$ so w/ FSDP + TP we can be compute-bounded.

$$X_{opt} = \sqrt{\frac{B}{F} \cdot \frac{M_x}{M_y} \cdot N}$$

choose $M_x = 2$, $M_y = 1$

$$X_{opt} = \sqrt{\frac{(4 \cdot 10^6)}{28672} \cdot 2 \cdot 8096} \simeq 1502$$

$\Rightarrow$ 2048-way FSDP and 4-way TP

Worked Problems:

1) $N = 4 \cdot 8960 = 35840$

When scaling up to multiple pods, we would use pure dp across the pods (we want minimal comm over dcn) and a different parallelism scheme within each pod depending on the new per pod batch size.

$BS_{local} = 4 \cdot 10^6 / 4 = 1 \cdot 10^6$

w/ pure TP in each pod:

$$\frac{F}{8960} = \frac{28672}{8960} < \frac{2550}{3} \quad \text{(comm-bounded)}$$

TP + FSDP:

$$\frac{B_{local}}{N} = 111 < \frac{2550^2}{2F} = 113$$

Between 4-pods, within each pod we are now comm bounded regardless of what parallelism scheme we choose.

In the intra-pod level, setting $M_x = 2, M_y = 1$:

$$X_{opt} = \sqrt{\frac{B_{local}}{F} \cdot \frac{M_x}{M_y} \cdot N} = \underbrace{\tfrac{1}{2}}_{} \cdot 1502 = 751$$

bc our batchsize decreases by $\frac{1}{4}$ which we pull out of the sqrt

$\Rightarrow$ Rounding to the nearest power of 2 gives

512-way FSDP and 16-way TP. (We have chips left over though)

At the inter-pod level:

$$T_{comm} = \frac{2 \cdot 2 \cdot 2 \cdot DF}{M \cdot W_{dcn}}$$

$$T_{math} = \frac{4 \cdot 2BDF}{N \cdot C}$$

$$T_{math} > T_{comm} \Rightarrow \frac{\cancel{8}BD\cancel{F}}{NC} > \frac{\cancel{8D F}}{MW_{dcn}}$$

$$\Rightarrow \frac{B \cdot M}{N} > \frac{C}{W_{dcn}}$$

$$\frac{\cancel{4} \cdot 10^6}{\cancel{4}} > 71{,}360$$

We will be compute-bounded over dcn.

Assuming MFU=1, looking at inter-node level and assuming perfect overlap of comp and comm intra-node. Also ignoring that we technically don't use all chips in above parallelism.

$$T_{Train} = \frac{6 \cdot 70B \cdot 15T}{4 \cdot 8960 \cdot 4.59 \cdot 10^{14}} = \frac{6 \cdot 70 \cdot 10^9 \cdot 15 \cdot 10^{12}}{4 \cdot 8960 \cdot 4.59 \cdot 10^{14}} \approx 3.8 \cdot 10^5 \, s$$

2) a) i)

hyperparam	value
n_layers (L)	126
d_model (D)	16384
d_{ff} (F)	53248
n_heads (N)	128

n_kv_heads (K)	8
d_qkv (H)	$D/N = 16384/128 = 128$
n_embeddings (V)	128 256

ii) Per layer:

$$\text{Attn} = DNH + DKH + DKH + NHD + D$$

$$= 2DNH + 2DKH + D \simeq 2DNH + 2DKH$$

$$\text{MLP} = 2DF + FD = 3DF$$

$$\text{Total} = (2DNH + 2DKH + 3DF) \cdot L + 2DV$$

$$= (2 \cdot 16384 \cdot 128 \cdot 128 + 2 \cdot 16384 \cdot 8 \cdot 128) \cdot 126$$

$$+ 3 \cdot 16384 \cdot 53248 + 2 \cdot 16384 \cdot 128256$$

$$= 4.05 \cdot 10^{11} = 405 \cdot 10^{9} \text{ params}$$

iii) $\text{FLOPS}/_{\text{Token}} = 6 \cdot 405 \cdot 10^9 = 2.43 \cdot 10^{12}$

iv) $\text{FLOPS}_{\text{Total}} = 2.43 \cdot 10^{12} \cdot 15 \cdot 10^{12} = 3.645 \cdot 10^{25}$

b) Assuming the actual sequence lengths are $< 8D$ so that attn costs are negligible and that our batch size is 4mil.

Across the pods I would use pure DP (minimize comm times over dcn):

$$\frac{B \cdot M}{N} = \frac{4 \cdot 10^6}{8} = 5 \cdot 10^5 > \frac{C}{W_{dcn}} = 73440$$

$\Rightarrow$ compute bound over dcn

At the intra-pod level:

Pure-DP will not work since model params alone are $405 \cdot 10^9 \cdot 2 = 810\,GB \gg 96\,GB$ HBM/vSp.

Trying pure FSDP:

$$\frac{B_{local}}{8960} = \frac{(4 \cdot 10^6/8)}{8960} \approx 55 < \frac{2550}{3} = 850$$

We can't use pure FSDP since we'll be comm-bound.

FSDP + TP (Let $M_x = 2$, $M_y = 1$):

$$\frac{B_{local}}{N} = \frac{(4 \cdot 10^6/8)}{8960} \approx 56$$

$$\frac{\alpha^2}{M_x M_y F} = \frac{2550^2}{2 \cdot 53248} \approx 61$$

We will still be slightly comm-bounded.

$$X_{opt} = \sqrt{\frac{B_{local}}{F} \cdot \frac{M_x}{M_y} \cdot N} = \sqrt{\frac{(4 \cdot 10^6/8)}{53248} \cdot 2 \cdot 8960} \approx 410$$

Closest power of 2 (for wraparound ici) gives us: 448-way FSDP and 20-way TP.

Assuming MFU = 1:

$$T_{Train} = \frac{6 \cdot 405 \cdot 10^{9} \cdot 15 \cdot 10^{12}}{8 \cdot 8960 \cdot 4.59 \cdot 10^{14}} \approx 1 \cdot 10^{6} s \approx 12 \text{ days}$$